

Abdullah Azhar

AI / ML Engineer

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SUMMARY

AI/ML engineer pursuing BS in Artificial Intelligence at the University of Faisalabad (CGPA 3.70), with hands-on production experience in computer vision, object detection, LLM fine-tuning, and full-stack development. Deployed real-world systems including a package label classifier (97% accuracy on Raspberry Pi 4) and a QLoRA fine-tuned Phi-3 Mini study assistant with FAISS-powered RAG. Contributed to Alzheimer's disease MRI classification research alongside a faculty PhD supervisor, achieving 95%+ accuracy across 4 severity classes. Active IEEE student branch member. Seeking ML/AI internship or junior engineer roles.

TECHNICAL SKILLS

- **Core Languages:** Python (primary), JavaScript, SQL, HTML5, CSS3
- **ML / AI:** PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, YOLOv8, Hugging Face Transformers, QLoRA, FAISS, LLaMA, GPT4All
- **Web Development:** React.js, Next.js, Node.js, Express.js, FastAPI, Flask, Streamlit, Gradio, MongoDB, Firebase, Supabase
- **Tools & Platforms:** Git, GitHub, REST APIs, Figma, VS Code, Raspberry Pi 4, Jetson Nano, Agile Development

WORK EXPERIENCE

Machine Learning Engineer | AI Predictions

Aug 2025 – Present

<https://aipredictions.ai/creators/9f7d6374-e1bf-48da-b199-a17a3195db5f/details>

- Built and deployed production-ready ML models for cryptocurrency and commodity price forecasting using ensemble methods.
- Published predictive models on the AI Predictions platform, serving live financial data for real-time decision-making tools.

Machine Learning Engineer | Visibility Bots

Jul 2025 – Oct 2025

- Contributed to a team building a GAN-based virtual try-on system (VITON-HD / PyTorch), responsible for model inference, testing, and bug fixing.
- Built a real-time fire and smoke detection model using TensorFlow and OpenCV, achieving 92% accuracy, deployed on Jetson Nano.
- Developed a package label prediction system achieving 97% accuracy (~1,074/1,100 samples), deployed on Raspberry Pi 4 for Kashmir LTD's factory production line.
- Built a Personal Protective Equipment (PPE) detection system using YOLOv8, deployed on Raspberry Pi for live factory floor safety monitoring.

Web Developer & Workshop Instructor | Youth Empowerment (MLSA)

Mar 2025 – Present

- Delivered certified workshops on JavaScript (beginner to advanced), then shifted to AI/ML: real-world project building and ML-into-web integration, training 50+ students total.
- Built responsive web applications for organisational initiatives using React.js, Next.js, and Tailwind CSS.

AI Intern | Digital Empowerment Network

Sep 2024 – Oct 2024

- Completed intensive C++ programming training, recognised for outstanding performance among the batch.
- Served as NSPBP Ambassador (ID: NSPBP 875), Dec 2024 to Aug 2025.

PROJECTS

Lecture-Grounded Study Assistant • Phi-3 Mini, QLoRA, FAISS, RAG, FastAPI, Streamlit

- Built and compared 3 model variants for lecture Q&A: base Phi-3 Mini (baseline), FAISS RAG-enhanced, and QLoRA fine-tuned (LoRA $r=16$, 3 epochs, SFT on slide-derived dataset via OpenRouter / Llama 3.1).
- End-to-end pipeline: PDF/PPTX slide parsing, FAISS vector indexing, FastAPI REST backend, Streamlit evaluation dashboard comparing all 3 models side-by-side.
- ANN & Deep Learning final project, University of Faisalabad — supervised by Dr. Muhammad Gufran Khan.

Alzheimer's Disease MRI Classification — PhD Research Contribution • *PyTorch, EfficientNet, ResNet, SMOTE, TTA*

- Contributed to faculty PhD research classifying Alzheimer's severity from MRI scans across 4 classes (No Dementia, Mild, Moderate, Very Mild) on a dataset of ~10,880 images.
- Designed dual-backbone feature extraction pipeline (4,608-dim fused features), applied SMOTE to correct severe class imbalance (Moderate class: only 88 training samples), and implemented 50x Test-Time Augmentation, reaching 95%+ test accuracy.

Package Label Prediction — Kashmir LTD • *Python, YOLOv8, Raspberry Pi 4*

- Trained and deployed an automated package label detection system for Kashmir LTD's factory production line, achieving 97% accuracy (~1,074/1,100 samples) on Raspberry Pi 4 edge hardware.

Real-Time Fire & Smoke Detection • *Python, TensorFlow, OpenCV, Jetson Nano*

- Trained a fire and smoke detection model achieving 92% accuracy, deployed on Jetson Nano for live environment monitoring at Visibility Bots.

PPE Detection System • *Python, YOLOv8, Raspberry Pi*

- Real-time Personal Protective Equipment detection deployed on Raspberry Pi for factory floor safety monitoring; detects helmet, vest, and glove compliance from live camera feed.

Shop People Counter • *Python, YOLOv8, SORT, OpenCV*

- Retail visitor counting system tracking people entering, exiting, and present simultaneously using multi-object SORT tracking on live CCTV feeds.

Personalized AI Assistant • *Python, LLaMA, GPT4All, Streamlit*

- Offline personal AI assistant running local LLMs via GPT4All; primarily terminal-based with optional Streamlit UI for visual interaction.

Virtual Try-On System • *VITON-HD, PyTorch, Gradio*

- Team project implementing GAN-based garment overlay for e-commerce visualisation; responsible for model inference pipeline, UI integration, testing, and bug fixing.

Object Detection System • *Python, YOLOv3, Flask*

- Full-stack web app serving YOLOv3 object detection inference results through a Flask backend and browser UI.

Predictive ML Models — BTC, Gold, Car Price • *Python, scikit-learn, Keras, Flask, Streamlit*

- Ensemble and regression models for financial forecasting and price prediction; BTC and Gold models selected and published on AI Predictions platform.

E-Commerce Platforms — Poshzem & EloraMate Gifts • *Next.js, MongoDB, Firebase, Supabase*

- Two full e-commerce platforms with product listings, authentication, and order management, with manual feature customisation on top of scaffolded codebase.

University Placement Bureau AI Integration • *Python, REST API*

- Integrated an AI recommendation model into the official TUF Placement Bureau website as Head Lead, improving student-to-job matching.

Architectural Diagram Generator • *Python, OpenCV, LLM*

- Automated extraction and reasoning over architectural diagrams by combining a computer vision pipeline with LLM-based structured output generation.

EDUCATION

BS Artificial Intelligence | [University of Faisalabad](#) 2023 – 2027
CGPA: 3.70

FSc Pre-Medical | [KIPS College, Faisalabad](#) 2020 – 2022
Grade A

Matriculation — Biological Sciences | [Govt. High School 117JB Dhanola](#) 2018 – 2020
Grade A+ (98%)

CERTIFICATIONS

- IEEE 3rd International Conference on Advanced STEAM — Certificate of Participation
- IEEE Certificate — New Technologies & AI Integration for Green Pakistan, Lahore Subsection
- IEEE Faisalabad Students, Young Professionals & Women in Engineering Congress — Certificate of Appreciation (Oct 2025)
- Discover AI in Daily Life — Google, 2025
- Building ML Models using Python — Microsoft Learn, 2024
- Diploma in Full Stack Development
- Diploma in Advanced Python

ACHIEVEMENTS & LEADERSHIP

- IEEE Student Branch Member — University of Faisalabad chapter.
- Google Developer Hackathon (Daira @ FAST University): Shortlisted in top 20 teams out of 60 based on proposed idea.
- AI Creators Platform: Selected to publish predictive models on the AI Predictions platform.
- NSPBP Ambassador (ID: NSPBP 875) — Digital Empowerment Network, Dec 2024 to Aug 2025.
- Workshop Instructor: Trained 50+ students across certified JavaScript and AI/ML workshops (MLSA).
- C++ Training Recognition: Outstanding performance award at Digital Empowerment Network internship.